



1st Place Solution: Multi-Iterative Noisy Student Is All You Need



BirdCLEF+ 2025

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I know that Kagglers are aware of what's happening in Ukraine, and you're here to take a look at my solution, but let me share a small glimpse from the competition's deadline day that reflects our current reality: I had to make my final submissions from a shelter while we had a stable connection, as Ukraine was under another devastating attack with many civilian casualties, some of them from my neighborhood.

I would like to thank the Armed Forces of Ukraine, Security Service of Ukraine, Defence Intelligence of Ukraine, and the State Emergency Service of Ukraine for providing safety and security to participate in this great competition, complete this work, and help science, technology, and business not to stop but to move forward.

TLDR

- SED models on 20-second input chunks.
- A Multi-Iterative Noisy Student is used as a self-training approach via MixUp between focal training data and pseudo-labeled soundscapes.
- Power transform applied to pseudo-labels to reduce noise.
- Pseudo-label sampler assigns weights equal to the sum of the maximum of labels within each soundscape.
- A separate model for Amphibia and Insecta label groups using extended species data from Xeno-Canto.
- A final ensemble with models from different training iterations.
- Inference is performed by averaging overlapping framewise predictions from neighboring chunks, followed by smoothing and delta shift inference.

Solution Overview